

An interpretable machine learning model for predicting prognosis of medulloblastoma integrating genetic and clinical features

Corresponding Author: Dr Yu Tian

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors present different models using multi-layer data from medulloblastoma cohorts (clinical, radiotherapy, molecular) to assess prognosis; Models were trained on 729 and validated in 201 patients.

The authors highlight integration of radiotherapy a key novel parameter in their analysis, and its importance w.r.t normal tissue toxicity and long term rt induced impairments. Thus, more detailed information about the irradiation technique should be provided/ commented on (i.e. irradiation with particles? hypofractionation? time of RT start following surgery? Boost -> local outcome vs evaluated overall survival?). Similarly, chemotherapy seems to be considered as dichotomous variable and should also be commented on more in depth (also in combination w. RT), as well as the extent of surgery (known predictor or residual tumor in early post-OP MRI).

As WNT group MB has a tremendously differently prognosis from SHH / Gr3 / Gr4 (an more frequently germline predisposition mutations), it might be worth to stratify analyses / test for interactions for the molecular subgroup.

Testing for baseline differences (Tbl. 1+2) using chi-squared tests seems odd for time-to-event data with censored observations; appropriate statistical tests should be used.

Reviewer #2

(Remarks to the Author)

This manuscript presents machine learning-based prognostic models for medulloblastoma, integrating both clinical and molecular features. The authors have also developed an interactive, web-based Shiny application to facilitate dynamic risk stratification and support treatment decision-making. The study has several strengths, including its relatively large sample size, the use of external validation, and the integration of multimodal data sources. However, I have several concerns regarding the methodology and study design, as outlined below:

1. Model selection and potential data leakage: The authors have tested multiple survival analysis approaches, including Cox proportional hazards (Cox-PH), DeepSurv, and XGBoost. However, it appears that all methods were evaluated on both the internal and external datasets without a clear separation of the model selection process from external testing. This approach risks model overfitting and information leakage, as the external dataset should be reserved strictly for final validation after model selection is complete. A more rigorous benchmarking framework is needed.

2. Internal data splitting for generalization: Given the large sample size available from the internal institution, it would have been preferable to split this dataset into a discovery (training/validation) set and an independent replication set that remained unseen during model training and hyperparameter tuning. Cross-validation should be applied only within the discovery set. This approach would help ensure better generalizability and reduce the risk of overfitting before testing on the external cohort.

3. Multi-institutional data harmonization: The analysis incorporates multi-institutional data, with molecular profiling performed

using different platforms (e.g., DNA methylation arrays, NGS, and RNA-seq). The manuscript does not clearly describe how the authors ensured compatibility and comparability of molecular results across these techniques. Standardization or harmonization steps should be detailed.

4. Batch effect correction: Relatedly, it is unclear how the authors accounted for potential batch effects arising from different assay platforms, laboratory protocols, or sequencing dates. Batch effect correction is critical in multi-platform molecular datasets, and the absence of these details raises concerns about confounding effects in the molecular predictors.

5. Impact of evolving therapeutic practices: As some cases in the dataset are more recent, they may have been treated under evolving therapeutic paradigms or clinical trial protocols. The authors should clarify how the influence of new therapies was accounted for in the modeling, as changes in standard of care could bias survival estimates.

6. Choice of survival endpoint: While overall survival (OS) is an important clinical endpoint, in medulloblastoma. As for relapsed cases, treatment options are limited, and the prognosis is poor, recurrence-free survival (RFS) or event-free survival (EFS) may be equally, if not more, clinically relevant for informing treatment planning and surveillance strategies. The authors should justify their choice of OS as the sole or primary endpoint and discuss whether their models could be extended to predict RFS/EFS.

7. Web application accessibility: The web links provided for accessing the publicly available Shiny app currently return an error. The authors should verify the functionality of these links and ensure the application is accessible for peer review and subsequent readership.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors have addressed most of the comments; however, several issues remain. In particular, the authors need to ensure that all comments are addressed directly within the manuscript and any supporting figures and tables are included in the main manuscript, rather than only in the rebuttal letter. This seems to be an issue for the responses to both reviewers' comments. Examples include Reviewer 1 Comment 1 & 2 (including response letter tables 1-4 & response letter figure 1), Reviewer 2 Comments 4-6 (including response letter figures 2-4).

As a specific example, I was unable to locate any results on disease-free survival as indicated in the rebuttal letter. Moreover, it is unclear where the figures and tables referenced in the rebuttal have been incorporated into the manuscript. Please include these results in the main manuscript where possible, or in the supplementary materials, and ensure that all figures and tables are appropriately cited in the text.

Also, the manuscript does not currently provide details on the patients' treatment strategies, despite this being requested in response to Reviewer 1, Comment 1, and Reviewer 2, Comment 5. These details need to be added to the manuscript.

Furthermore, the authors should temper their claims regarding the performance of their models. Describing the models as "excellent," for example, is not supported by the reported metrics; the performance appears modest at best.

3.

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

Thank you for addressing the comments.

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Dear Editors,

We really appreciate the editorial board members, and reviewers for the valuable comments and suggestions, which greatly improve our manuscript. We have revised the manuscript accordingly and responded to comments point by point in the response letter (enclosed). All changes in the revised manuscript have been highlighted in yellow. We believe that the quality of manuscript has been greatly improved after revision.

Thank you again for your consideration and support!

Yours sincerely,

Yu Tian, on behalf of all co-authors

Reviewer(s)' Comments to Author:

Reviewer 1

The authors present different models using multi-layer data from medulloblastoma cohorts (clinical, radiotherapy, molecular) to assess prognosis; Models were trained on 729 and validated in 201 patients.

* 1. The authors highlight integration of radiotherapy a key novel parameter in their analysis, and its importance w.r.t normal tissue toxicity and long term rt induced impairments. Thus, more detailed information about the irradiation technique should be provided/ commented on (i.e. irradiation with particles? hypofractionation? time of RT start following surgery? Boost -> local outcome vs evaluated overall survival?). Similarly, chemotherapy seems to be considered as dichotomous variable and should also be commented on more in depth (also in combination w. RT), as well as the extent of surgery (known predictor or residual tumor in early post-OP MRI).

We thank the reviewer for the constructive comments and suggestions. Reviewer 2 also focused on this point in question 5. All patients received X-ray photon radiotherapy delivered with conventional fractionation (1.8 Gy per fraction) in accordance with the NCCN guidelines. Intensity-modulated radiotherapy (IMRT) was used before 2012, and volumetric-modulated arc therapy (VMAT) was gradually adopted thereafter. The median interval between surgery and initiation of radiotherapy was 38 days, and the median duration of radiotherapy was 42 days. No concurrent chemotherapy was administered during radiotherapy. The median CSI dose was 30.6 Gy, and the median posterior fossa/tumor bed (PFTB) boost dose was 55.8 Gy. Adjuvant chemotherapy was initiated 4–6 weeks after completion of radiotherapy, with regimens determined according to risk stratification. The extent of

resection was assessed by contrast-enhanced MRI within 72 hours after surgery, and the gross-total resection rate in this cohort was 91.2%. The treatment strategies for RT and RT+CT have been incorporated and accounted for in the two clinical scenarios analyzed in our modeling.

*2. As WNT group MB has a tremendously different prognosis from SHH / Gr3 / Gr4 (an more frequently germline predisposition mutations), it might be worth to stratify analyses / test for interactions for the molecular subgroup.

We thank the reviewer for raising this important point regarding the differences between WNT-MB and other molecular subgroups. We included the interaction terms in the multivariate Cox analysis. There was a significant interaction between molecular subgroup and CSI dose on the outcome (Response letter table 1). In contrast, as shown in Tables 2 and 3 of the response letter, no significant interaction was found when PFTB boost dose, and treatment were analyzed separately for their interactions with the molecular subgroup. We observed no deaths in the WNT subgroup when the CSI doses were ≥ 30.6 Gy. Therefore, we expect a more substantial number of cases in the WNT subgroup at the CSI doses in the further follow-up or future studies to supplement the existing evidence.

We have discussed the impact of risk stratification across different subtypes and treatment strategies on patient survival in another paper, titled “Long-term outcomes by clinical and molecular risk stratification in medulloblastoma patients receiving risk-adapted therapy” (Response letter figure 1 and table 3). This manuscript is now in the second round of review after minor revision. Molecularly, WNT-MB patients had better prognostic benefits than the other three subgroups (Response letter figure 1). However, no statistically significant differences were observed in extent of resection, CSI dose, treatment strategies, or PFTB boost dose distribution between the WNT subgroup and the remaining subgroups (Response letter table 4). Therefore, patients from all four subgroups were collectively included in the model construction.

Response letter table 1. Multivariate Cox Analysis of interaction term (Subgroup and CSI dose).

Characteristic	HR (95% CI)	p-value
Subgroup		
WNT	Ref	-
Non-WNT	1.26 (0.44-3.62)	0.67
CSI dose		
[0-30.6)	Ref	-
≥ 30.6	4.20e-08 (0.0000-Inf)	0.99

Subgroup * CSI dose	-	0.01
RERI	-	<0.01

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio, RERI = Relative excess risk due to interaction. Adjusted for Sex, Age, M stage, Histology, Excision, Treatment, PFTB boost dose.

Response letter table 2. Multivariate Cox Analysis of interaction term (Subgroup and PFTB boost dose).

Characteristic	HR (95% CI)	p-value
Subgroup		
WNT	Ref	-
Non-WNT	3.20 (0.42-24.15)	0.26
PFTB boost dose		
<55.8	Ref	-
≥55.8	1.10 (0.11-10.82)	0.94
Subgroup * PFTB boost dose	-	0.88
RERI	-	0.59

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio, RERI = Relative excess risk due to interaction. Adjusted for Sex, Age, M stage, Histology, Excision, Treatment, CSI dose.

Response letter table 3. Multivariate Cox Analysis of interaction term (Subgroup and Treatment).

Characteristic	HR (95% CI)	p-value
Subgroup		
WNT	Ref	-
Non-WNT	2.05 (0.48- 8.79)	0.33
Treatment		
RT	Ref	-

RT+CT	0.40 (0.05- 2.87)	0.36
Subgroup * Treatment	-	0.60
RERI	-	0.52

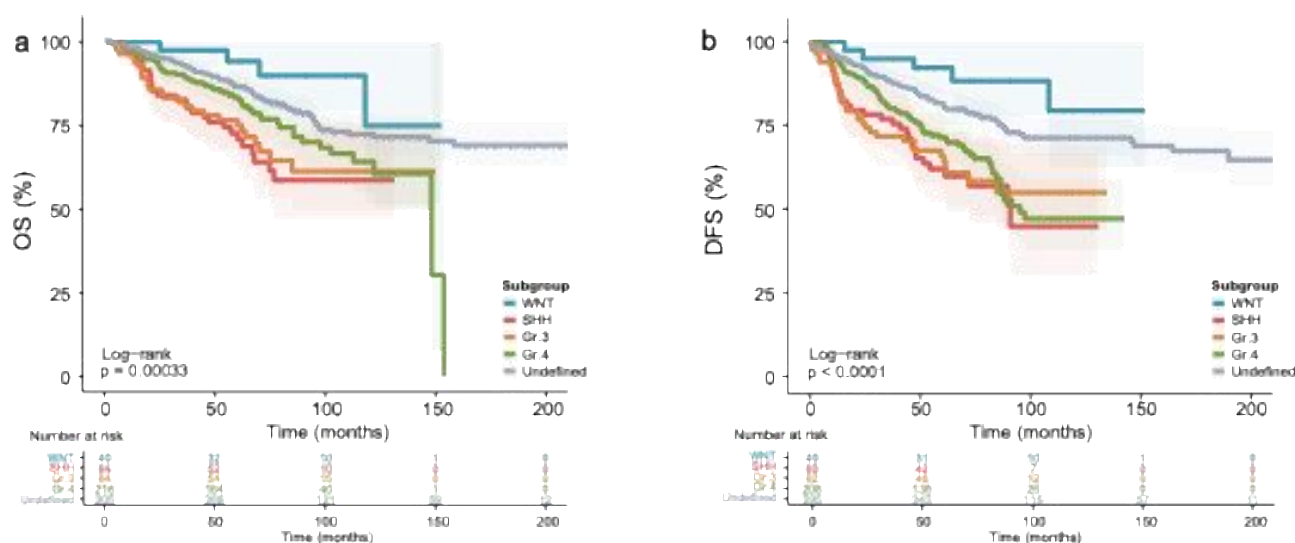
Abbreviations: CI = Confidence Interval, HR = Hazard Ratio, RERI = Relative excess risk due to interaction. Adjusted for Sex, Age, M stage, Histology, Excision, CSI dose, PFTB boost dose.

Response letter table 4. Demographic and baseline clinicopathological characteristics of molecularly defined cases (N = 424)

Variable	Overall (N = 424) n (%)	Non-WNT N = 384 n (%)	WNT N = 40 n (%)	p-value ¹
Sex				0.149
Female	147 (34.67%)	129 (33.59%)	18 (45.00%)	
Male	277 (65.33%)	255 (66.41%)	22 (55.00%)	
Age				0.191
Infant	40 (9.43%)	39 (10.16%)	1 (2.50%)	
Child	287 (67.69%)	261 (67.97%)	26 (65.00%)	
Adolescent	68 (16.04%)	58 (15.10%)	10 (25.00%)	
Adult	29 (6.84%)	26 (6.77%)	3 (7.50%)	
M stage				0.071
M0	348 (82.08%)	311 (80.99%)	37 (92.50%)	
M+	76 (17.92%)	73 (19.01%)	3 (7.50%)	
Histology				<0.001
CMB	215 (50.71%)	182 (47.40%)	33 (82.50%)	
DNMB	115 (27.12%)	111 (28.91%)	4 (10.00%)	
MBEN	22 (5.19%)	22 (5.73%)	0 (0.00%)	
LC/AMB	29 (6.84%)	29 (7.55%)	0 (0.00%)	
Undefined	43 (10.14%)	40 (10.42%)	3 (7.50%)	
Excision				0.291

STR	48 (11.32%)	46 (11.98%)	2 (5.00%)
GTR/NTR	376 (88.68%)	338 (88.02%)	38 (95%)
Treatment			0.537
RT	80 (18.87%)	71 (18.49%)	9 (22.50%)
RT+CT	344 (81.13%)	313 (81.51%)	31 (77.50%)
CSI dose			0.245
[0-30.6)	145 (34.20%)	128 (33.33%)	17 (42.50%)
≥30.6	279 (65.80%)	256 (66.67 %)	23 (57.50 %)
PFTB boost			0.676
<55.8	107 (25.24%)	98 (25.52%)	9 (22.50%)
≥55.8	317 (74.76%)	286 (74.48%)	31 (77.50%)

¹Pearson's Chi-squared test or Fisher's exact test



Response letter figure 1. Kaplan-Meier plot showing OS (a) and DFS (b) for the molecular subgroups.

*3. Testing for baseline differences (Tbl. 1+2) using chi-squared tests seems odd for time-to-event data with censored observations; appropriate statistical tests should be used.

Thanks for the reviewer's kindly reminding. We sincerely appreciate your pointing out the inappropriate use of chi-squared tests in the original Tables 1 and 2 for analyzing time-to-event data with censored observations.

In response to this, we have revised Tables 1 and 2. Now, they present the composition descriptions of the training set, testing set, and external validation set. This adjustment aims to provide a more accurate and appropriate overview of the data structure.

Reviewer 2

This manuscript presents machine learning-based prognostic models for medulloblastoma, integrating both clinical and molecular features. The authors have also developed an interactive, web-based Shiny application to facilitate dynamic risk stratification and support treatment decision-making. The study has several strengths, including its relatively large sample size, the use of external validation, and the integration of multimodal data sources. However, I have several concerns regarding the methodology and study design, as outlined below:

*1. Model selection and potential data leakage: The authors have tested multiple survival analysis approaches, including Cox proportional hazards (Cox-PH), DeepSurv, and XGBoost. However, it appears that all methods were evaluated on both the internal and external datasets without a clear separation of the model selection process from external testing. This approach risks model overfitting and information leakage, as the external dataset should be reserved strictly for final validation after model selection is complete. A more rigorous benchmarking framework is needed.

We sincerely thank you for your valuable comments. To address these concerns, we divided the data from the Chinese cohort, with 70% utilized for training and 30% for testing (internal validation), to avoid problems with overfitting and potential data leakage. In addition, an external data set was used for validation (external validation). We have added the following information in the Method section of the revised manuscript:

The data from the Chinese cohort were divided, with 70% utilized for training and 30% for testing, in order to avoid overfitting. Additionally, an external international set was used for external validation.

*2. Internal data splitting for generalization: Given the large sample size available from the internal institution, it would have been preferable to split this dataset into a discovery (training/validation) set and an independent replication set that remained unseen during model training and hyperparameter tuning. Cross-validation should be applied only within the discovery set. This approach would help ensure better generalizability and reduce the risk of overfitting before testing on the external cohort.

We sincerely thank the reviewer for your valuable comments. To address your concerns about model selection and potential data leakage, and consistent with our response to the previous question, we use a strict data-splitting and validation approach:

First, we divided the Chinese cohort data into a 70% training set and a 30% testing set. We did 5-fold cross-validation on the training set to tune the models' hyperparameters. The best model was chosen based on performance results from the independent testing set—importantly, this testing set was not used at all during model training or hyperparameter tuning. This effectively reduced the risks of overfitting and data leakage.

Additionally, to make our final selected model more generalizable, we used an independent external dataset only for final validation. This external cohort was not involved in any step of model development or selection before validation.

We appreciate your focus on the rigor of this method, as it has helped us clarify and strengthen the reliability of our analysis framework.

*3. Multi-institutional data harmonization: The analysis incorporates multi-institutional data, with molecular profiling performed using different platforms (e.g., DNA methylation arrays, NGS, and RNA-seq). The manuscript does not clearly describe how the authors ensured compatibility and comparability of molecular results across these techniques. Standardization or harmonization steps should be detailed.

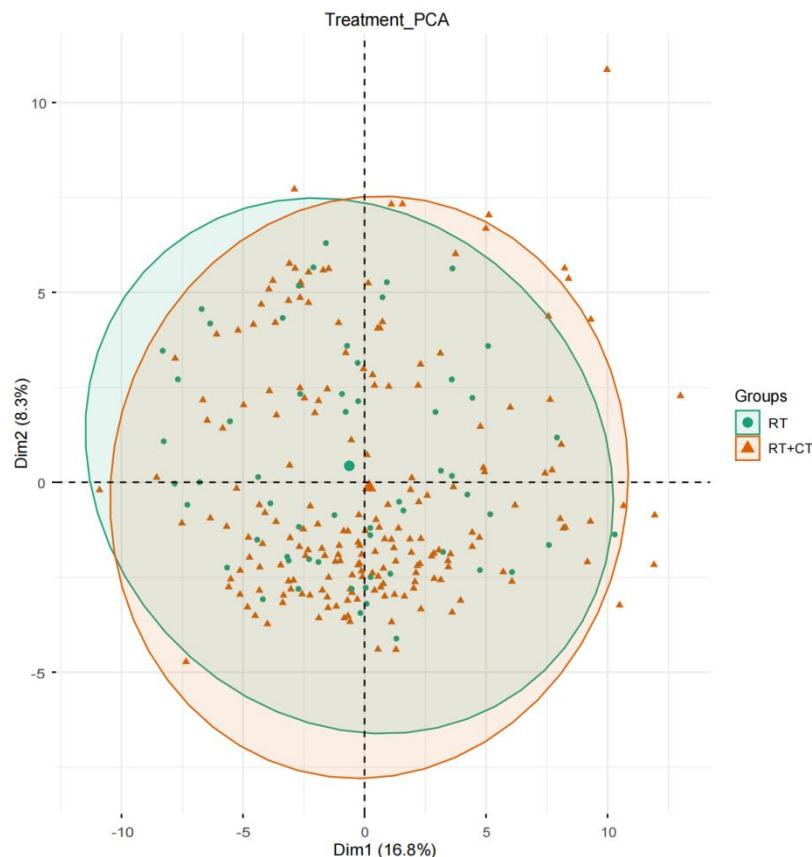
We appreciate the reviewer's attention to multi-institutional data harmonization. For the Chinese cohort, all molecular profiling (DNA methylation arrays, NGS, and RNA-seq) underwent standardized library construction and sequencing at Beijing Genomics Institute (BGI Research, Chongqing), ensuring consistent technical quality. This centralized processing strategy minimizes technical variability across platforms, enhancing the comparability of molecular results. We have added the following information in the Method section of the revised manuscript:

For the Chinese cohort, all molecular profiling (DNA methylation arrays, NGS, and RNA-seq) underwent standardized library construction and sequencing at Beijing Genomics Institute (BGI Research, Chongqing), ensuring consistent technical quality.

*4. Batch effect correction: Relatedly, it is unclear how the authors accounted for potential batch effects arising from different assay platforms, laboratory protocols, or sequencing dates. Batch effect correction is critical in multi-platform molecular datasets, and the absence of these details raises concerns about confounding effects in the molecular predictors.

Thanks for the reviewer's suggestions. For the International cohort, we apologize for the confusion. Although we used public data from multiple EGA IDs, all the samples were from MAGIC cohort, where all the library construction and sequencing were performed in the BC Cancer Agency's Michael Smith Genome Sciences Centre (BCGSC) with the same protocol ^[1] as previously described ^[2, 3]. For the Chinese cohort, all molecular profiling underwent standardized library construction and

sequencing at BGI, ensuring consistent technical quality. Before 2010, radiotherapy was the primary treatment approach. Following 2010, the standard of care shifted predominantly to a combination of radiotherapy and chemotherapy. Given the potential time-related effects of this treatment transition, we incorporated treatment strategy as a factor when evaluating batch effects via principal component analysis (PCA). The PCA results showed samples clustered tightly, with no distinct separation related to assay platforms, protocols, or sequencing dates—confirming no obvious batch effects (Response letter figure 2). This ensures no batch-related confounding in the molecular predictors.



Response letter figure 2. Principal component analysis (PCA) in Chinese cohort.

*5. Impact of evolving therapeutic practices: As some cases in the dataset are more recent, they may have been treated under evolving therapeutic paradigms or clinical trial protocols. The authors should clarify how the influence of new therapies was accounted for in the modeling, as changes in standard of care could bias survival estimates.

We sincerely thank the reviewer for the valuable suggestions. Reviewer 1 also focused on this point in question 1. All patients in the Chinese cohort received X-ray photon radiotherapy delivered with conventional fractionation (1.8 Gy per fraction) in accordance with the NCCN guidelines. Intensity-modulated radiotherapy (IMRT) was used before 2012, and volumetric-modulated arc therapy (VMAT) was gradually adopted thereafter. The median interval between surgery and initiation of

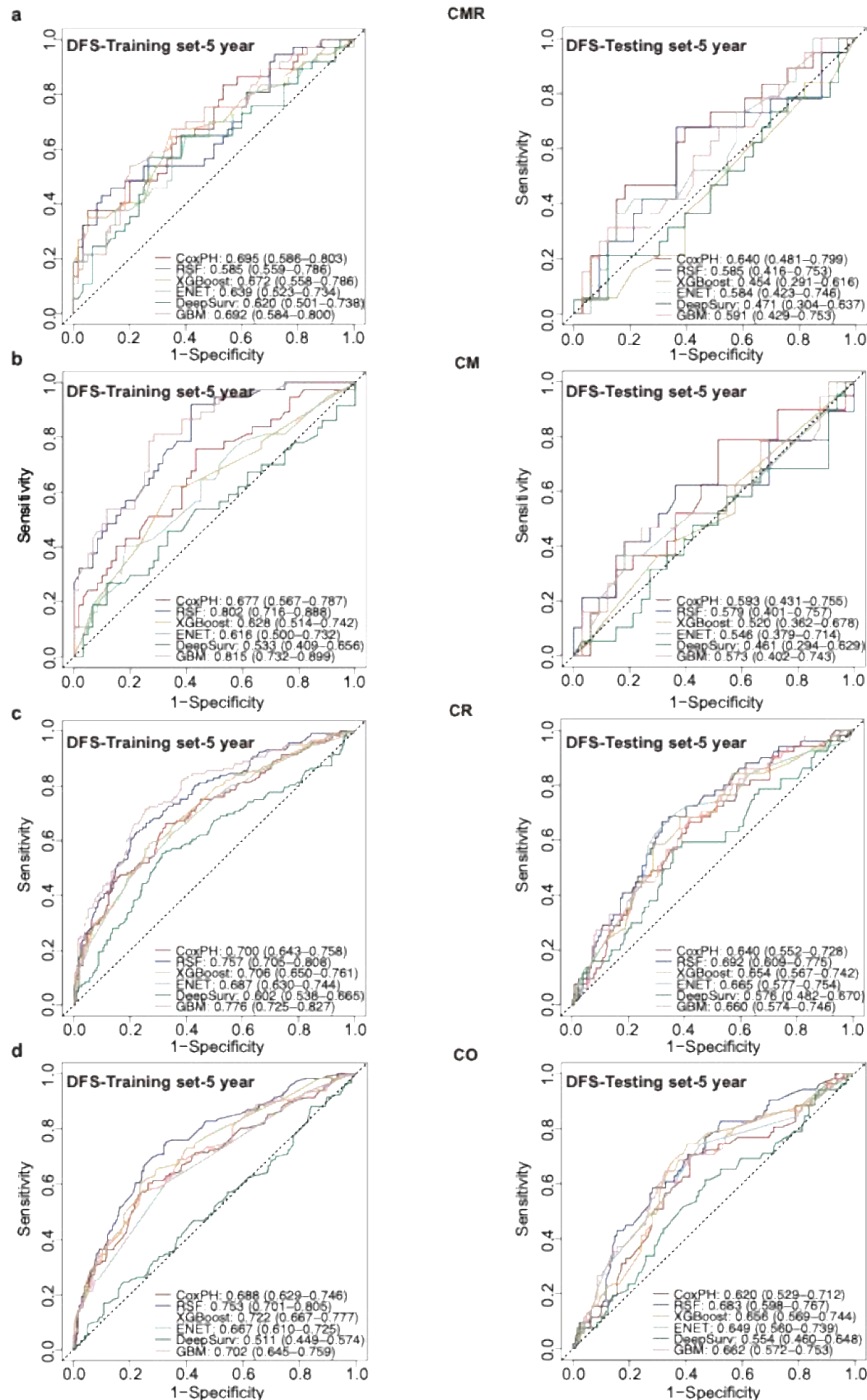
radiotherapy was 38 days, and the median duration of radiotherapy was 42 days. No concurrent chemotherapy was administered during radiotherapy. The median CSI dose was 30.6 Gy, and the median posterior fossa/tumor bed (PFTB) boost dose was 55.8 Gy. Adjuvant chemotherapy was initiated 4–6 weeks after completion of radiotherapy, with regimens determined according to risk stratification. The extent of resection was assessed by contrast-enhanced MRI within 72 hours after surgery, and the gross-total resection rate in this cohort was 91.2%. Our study focused exclusively on primary patients and did not analyze re-treatment after recurrence (the stage where new therapies might be involved). We further accounted for changes in treatment strategies: before 2010, radiotherapy alone was the standard of care, while after 2010, combined radiotherapy and chemotherapy became the standard. This difference in treatment strategies has been incorporated and accounted for in the two clinical scenarios analyzed in our modeling.

*6. Choice of survival endpoint: While overall survival (OS) is an important clinical endpoint, in medulloblastoma. As for relapsed cases, treatment options are limited, and the prognosis is poor, recurrence-free survival (RFS) or event-free survival (EFS) may be equally, if not more, clinically relevant for informing treatment planning and surveillance strategies. The authors should justify their choice of OS as the sole or primary endpoint and discuss whether their models could be extended to predict RFS/EFS.

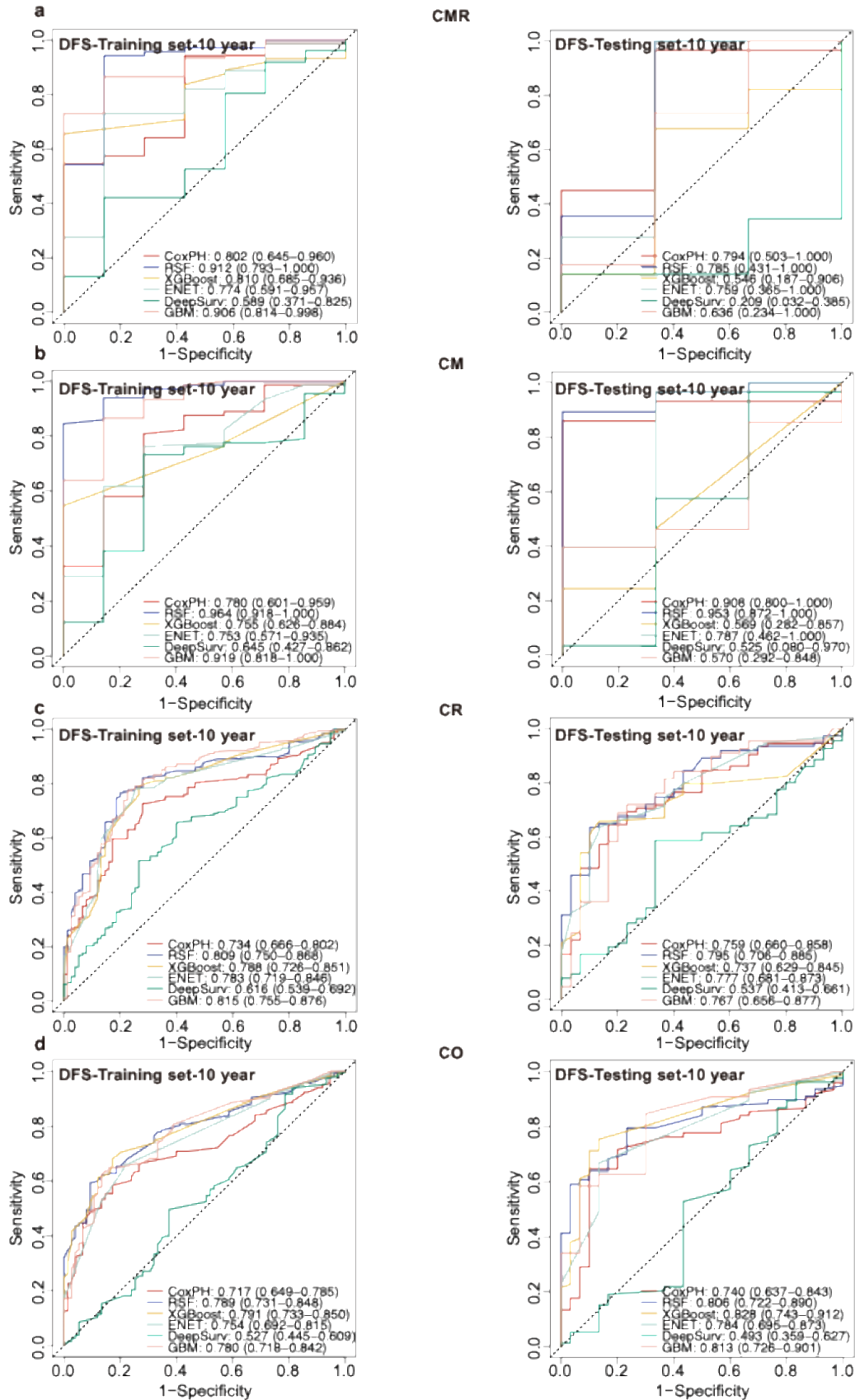
Thanks for the reviewer's suggestions. We sincerely appreciate your insightful comment on the survival endpoint. Basing on the available recurrence and mortality data in Chinese cohort, we analyzed disease free survival (DFS), defined as the period from treatment initiation to the first recurrence or death.

We developed four DFS prediction scenarios and assessed them in the testing set: (1) a CoxPH-based CMR model with a C - index of 0.657; (2) a GBM-based CM model with a C - index of 0.632; (3) a RSF-based CR model with a C - index of 0.659; (4) a RSF-based CO model with a C - index of 0.643 (Response letter figure 3, 4).

Unfortunately, external validation of the DFS models was unfeasible due to the lack of recurrence data in the external cohort. We intend to gather recurrence data to extend the external validation, thereby enhancing the clinical value of the models.



Response letter figure 3. ROC curves with different algorithms for predicting 5-year survival predictions in medulloblastoma patients in four scenarios.



Response letter figure 4. ROC curves with different algorithms for predicting 10-year survival predictions in medulloblastoma patients in four scenarios.

*7. Web application accessibility: The web links provided for accessing the publicly available Shiny app currently return an error. The authors should verify the functionality of these links and ensure the application is accessible for peer review and subsequent readership.

Thanks for the reviewer's suggestions. We sincerely apologize for the inconvenience caused by the error when accessing the Shiny app web links previously. The web applications for the four scenarios (CMR, CM, CR, and CO) have been successfully deployed, and their respective access links are provided below:

For the CMR scenario (applicable when radiotherapeutic dose information is available): https://prognosticmodel.shinyapps.io/Scenario1_CMR/

For the CM scenario (applicable when radiotherapeutic dose information is unavailable): https://prognosticmodel.shinyapps.io/Scenario2_CM/

For the CR scenario: https://prognosticmodel.shinyapps.io/Scenario3_CR/

For the CO scenario: https://prognosticmodel.shinyapps.io/Scenario4_CO/

We recommend accessing these links using a stable network connection. Should you encounter any persistent errors or accessibility issues, please contact us at your convenience. We will promptly provide assistance to resolve the issue, ensuring the applications remain accessible for review and future reference.

REF:

- [1] Morrissy A S, Garzia L, Shih D J, et al. Divergent clonal selection dominates medulloblastoma at recurrence [J]. Nature, 2016, 529(7586): 351-7.
- [2] Skowron P, Farooq H, Cavalli F M G, et al. The transcriptional landscape of Shh medulloblastoma [J]. Nat Commun, 2021, 12(1): 1749.
- [3] Hendrikse L D, Haldipur P, Saulnier O, et al. Failure of human rhombic lip differentiation underlies medulloblastoma formation [J]. Nature, 2022, 609(7929): 1021-8.

Dear Editors,

Happy new year!

We really appreciate the editorial board members, and reviewers for the valuable comments and suggestions, which greatly improve our manuscript. We have revised the manuscript accordingly and responded to comments point by point in the response letter (enclosed). The responses to the reviewers are provided herein, while the responses to the editors have been listed separately. All changes in the revised manuscript have been highlighted in yellow. We believe that the quality of manuscript has been greatly improved after revision.

Thank you again for your consideration and support!

Yours sincerely,

Yu Tian, on behalf of all co-authors

Reviewers' comments:

Reviewer #2 (Remarks to the Author):

1.The authors have addressed most of the comments; however, several issues remain. In particular, the authors need to ensure that all comments are addressed directly within the manuscript and any supporting figures and tables are included in the main manuscript, rather than only in the rebuttal letter. This seems to be an issue for the responses to both reviewers' comments. Examples include Reviewer 1 Comment 1& 2 (including response letter tables 1-4 & response letter figure 1), Reviewer 2 Comments 4-6 (including response letter figures 2-4).

Response:

We thank the Editor for the constructive feedback. Following the editor's suggestions, we have ensured that all responses and supplementary data previously provided in the rebuttal letter are now formally integrated into the revised manuscript or the supplementary material. Specific locations of these updates are detailed in our point-by-point responses below.

For Reviewer#1, Comment 1 in the previous letter “The authors highlight integration of radiotherapy a key novel parameter in their analysis, and its importance w.r.t normal tissue toxicity and long term rt induced impairments. Thus, more detailed information about the irradiation technique should be provided/ commented on (i.e. irradiation with particles? hypofractionation? time of RT start following surgery? Boost -> local outcome vs evaluated overall survival?). Similarly, chemotherapy seems to be considered as dichotomous variable and should also be commented on more in depth (also in combination w. RT), as well as the extent of surgery (known predictor or residual tumor in early post-OP MRI).”

Response:

We have added the following comprehensive description of the patients' treatment strategies in the Methods section of the revised manuscript (Line 162–180):

“Regarding treatment protocols, the extent of resection was assessed by contrast-enhanced MRI within 72 hours post-surgery, with a gross-total resection rate of 91.2% in this cohort. All patients received X-ray photon radiotherapy delivered with conventional fractionation (1.8 Gy per fraction) in accordance with NCCN guidelines. Intensity-modulated radiotherapy (IMRT) was predominantly used before 2012, whereas volumetric-modulated arc therapy (VMAT) was gradually adopted thereafter. The median interval between surgery and the initiation of radiotherapy was 38 days, and the median radiotherapy duration was 42 days. The median craniospinal irradiation (CSI) dose was 30.6 Gy, and the median posterior

fossa/tumor bed (PFTB) boost dose was 55.8 Gy. No concurrent chemotherapy was administered during radiotherapy. Adjuvant chemotherapy was initiated 4 – 6 weeks after the completion of radiotherapy, with regimens determined according to risk stratification. Before 2010, radiotherapy was the primary treatment approach. Following 2010, the standard of care shifted predominantly to a combination of radiotherapy and chemotherapy. Given the potential time-related effects of this treatment transition, we incorporated treatment strategy as a factor when evaluating batch effects via principal component analysis (PCA). The PCA results showed samples clustered tightly, with no distinct separation related to assay platforms, protocols, or sequencing dates—confirming no obvious batch effects (Supplementary Figure 1). This ensures no batch-related confounding in the molecular predictors. Therefore, the treatment strategies for RT and RT+CT were accounted for in the two clinical scenarios analyzed in our modeling.”

Reviewer#1, Comment 2 in the previous letter “As WNT group MB has a tremendously differently prognosis from SHH / Gr3 / Gr4 (an more frequently germline predisposition mutations), it might be worth to stratify analyses / test for interactions for the molecular subgroup.”

Response:

Thank you for this insightful comment. In the revised manuscript, we have performed a multivariable Cox analysis that incorporates interaction terms. This analysis and its interpretation have been added to the "Model performance comparison" section (Line 356–360): “Given the significant differences between WNT-MB and other molecular subgroups, we included interaction terms in the multivariate Cox analysis. As shown in Supplementary Tables 3 and 4, no significant interactions were found when PFTB boost dose and treatment strategies were analyzed separately with molecular subgroups. Therefore, we combined WNT with other subgroups to construct the predictive model.”

Furthermore, the detailed statistical results are now presented in revised Supplementary Tables 3 and 4.

Reviewer#2, Comment 4 in the previous letter “Batch effect correction: Relatedly, it is unclear how the authors accounted for potential batch effects arising from different assay platforms, laboratory protocols, or sequencing dates. Batch effect correction is critical in

multi-platform molecular datasets, and the absence of these details raises concerns about confounding effects in the molecular predictors”.

Response:

As requested, we have included the PCA (Principal Component Analysis) plot to illustrate the adjustment for batch effects. This figure has been added as revised Supplementary Figure 1 in the revised supporting information.

Reviewer 2, Comment 5 in the previous letter “Impact of evolving therapeutic practices:

As some cases in the dataset are more recent, they may have been treated under evolving therapeutic paradigms or clinical trial protocols. The authors should clarify how the influence of new therapies was accounted for in the modeling, as changes in standard of care could bias survival estimates.”

Response:

We have expanded the description of treatment modalities to provide more detail. These additions can be found in the Methods section of the revised manuscript (Line 162–180).

Reviewer#2, Comment 6 in the previous letter “Choice of survival endpoint: While overall survival (OS) is an important clinical endpoint, in medulloblastoma. As for relapsed cases, treatment options are limited, and the prognosis is poor, recurrence-free survival (RFS) or event-free survival (EFS) may be equally, if not more, clinically relevant for informing treatment planning and surveillance strategies. The authors should justify their choice of OS as the sole or primary endpoint and discuss whether their models could be extended to predict RFS/EFS.”

Response:

To address the reviewer’s request for DFS analysis, we have added a new subsection titled “Predictive Performance of Disease-Free Survival (DFS) Models” in the Results section (Line 389–398). The corresponding analysis results and survival curves have been included as revised Supplementary Figure 6.

2.As a specific example, I was unable to locate any results on disease-free survival as indicated in the rebuttal letter. Moreover, it is unclear where the figures and tables referenced in the rebuttal have been incorporated into the manuscript. Please include these results in the main manuscript where possible,

or in the supplementary materials, and ensure that all figures and tables are appropriately cited in the text.

Response:

We sincerely apologize for our oversight in the previous revision. We thank the Editor for pointing out that the disease-free survival (DFS) results discussed in our rebuttal were not properly incorporated into the manuscript text.

We have corrected this immediately. In the revised manuscript, we have added a dedicated subsection titled "Predictive Performance of Disease-Free Survival (DFS) Models" within the Results section (Line 389–398). The detailed analysis results and the corresponding survival curves have been included as revised Supplementary Figure 6, which is now appropriately cited within this newly added text.

We have also conducted a thorough check to ensure all other figures and tables mentioned in our response are correctly included and cited in the revised manuscript or supplementary materials.

3.Also, the manuscript does not currently provide details on the patients' treatment strategies, despite this being requested in response to Reviewer 1, Comment 1, and Reviewer 2, Comment 5. These details need to be added to the manuscript.

Response:

We appreciate the reviewer's insightful comment, which was also raised by the Comment 1. We have added a comprehensive description of the patients' treatment strategies in the Methods section of the revised manuscript (Line 162–180).

4.Furthermore, the authors should temper their claims regarding the performance of their models. Describing the models as "excellent," for example, is not supported by the reported metrics; the performance appears modest at best.

Response:

We appreciate the reviewer's feedback and have revised the manuscript to tone down overly enthusiastic language. Specifically, we have replaced subjective descriptors such as "excellent" with more measured and accurate characterizations based on the reported metrics. We have carefully revised the manuscript to moderate the strength of our statement during the revision process.

For example:

1. We change the sentence “The comparative performances of these six algorithms in four scenarios were summarized in Supplementary Table 2, demonstrating that in different scenarios, the prediction of XGBoost and GBM algorithms performed excellently in terms of discrimination and calibration across different scenarios” into “Their comparative performances are summarized in Supplementary Table 6, demonstrating that the XGBoost and GBM algorithms exhibited moderate yet consistent discrimination and favorable calibration across different scenarios” (Line 363–365).

2. We change the sentence “Collectively, XGBoost and GBM algorithm in the four scenarios demonstrated excellent calibration and generalizability in external validations.” into “Collectively, the XGBoost and GBM algorithms in the four scenarios showed favorable calibration and consistent performance in external validations.” (Line 418–419).